

Discussion of
“Interest Rate Risk and Bank Hedging”
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How Much Should Agents Hedge Risk ϵ ?

How **Does** ϵ Affect Agents Absent Hedging?

How **Should** ϵ Affect Agents **Optimally**?

How Much Should **Firms** Hedge Risk ϵ ?

How **Does** ϵ Affect Firms Absent Hedging?

$$E_0 \left[\sum_{t=0}^{\tau} M_t A_t \frac{D_t}{A_t} \right]$$

- Firm survival τ
- Cash flows
 - Firm size A_t (growth rate)
 - “Profitability” D_t/A_t
- Discount rates M_t
 - Frictionless world: shareholders $M_t = M_t^H$
 - Frictions: $M_t = M_t^H X_t$
 - E.g., equity issuance costs, borrowing constraints

How **Should** ϵ Affect Firms **Optimally**?

- Frictionless world: who cares!
 - Idiosyncratic shocks wash out in diversified portfolios
 - Shareholders can do their own risk management of systematic shocks
 - Firm hedging is then an agency conflict (Stulz 1984)
- Frictions
 - $X_t \neq 1$: Dollar inside firm $\neq$ dollar outside
 - Goal of firm risk management:
 - Stabilize some function of X_t
 - Froot, Scharfstein, and Stein 1993: *“If external sources of finance are more costly to corporations than internally generated funds, there will typically be a benefit to hedging: hedging adds value to the extent that it helps ensure that a corporation has sufficient internal funds available to take advantage of attractive investment opportunities.”*

“However, for banks, there is a further wrinkle. Because the discount rate for future cash flows is itself the interest rate, changes in interest rates simultaneously affect both the cash flows and their valuation [...] This dual sensitivity introduces additional complexity in determining the optimal hedging policy for value-maximizing banks.”

How is Bank of America's hedging of interest rate risk different from Delta's hedging of oil risk?

"However, for banks, there is a further wrinkle. Because the discount rate for future cash flows is itself the interest rate, changes in interest rates simultaneously affect both the cash flows and their valuation [...] This dual sensitivity introduces additional complexity in determining the optimal hedging policy for value-maximizing banks."

What Makes Banks Special? Candidates:

- Franchise value
 - Plenty of firms, including banks, have asset-side franchise value
 - Banks can generate rents on the liability side, but is that qualitatively important?
- High maturity mismatch + high leverage
 - Uniquely large exposure to shocks that mainly move discount rates
 - But rents may have opposite duration
- Regulatory environment distorts **cost-of-capital passthrough**
 - Deposit insurance makes debt too cheap
 - Capital regulation occasionally makes debt infinitely expensive
 - Barriers to entry allow large rents to be sustained in equilibrium

What Makes Interest Rate Risk Special?

- Moves both cash flows and discount rates
- But any systematic risk will do that
 - Risk-free rates respond to consumption smoothing, precautionary motives
 - Risk premia respond to changes in price and quantity of risk
- Insight: the “classical” treatment of firm hedging is for shocks that affect the firm but not its shareholders
- If Delta’s investors drive gas-powered cars, this paper is also for them!
- More broadly: what *is* interest rate risk?

This Paper

- To classic Corporate Finance hedging literature
 - Brings Aggregate [Interest Rate] Risk and [Intermediary] Asset Pricing
 - To recent literature on bank interest rate management
 - Brings a comprehensive, dynamic approach
 - Not just NIMs
 - Not just Asset + Liability Duration Matching
 - Asset growth rates, deposit growth rates, compounding, cash flow exposures to rate shocks, regulatory constraints, ...
 - ~~• Interaction with liquidity management~~
 - State-of-the-art continuous time finance model
 - Despite a small state space, tricky optimal control problem requires solving using neural nets!
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Hedging as Wedge Management

- One-period interest rate swap with no upfront cost
 - Fixed: k_t
 - Floating: r_{t+1}
 - k_t set by suppliers of swaps such that $0 = E_t[M_{t+1}(r_{t+1} - k_t)]$
- Banks with SDF $M_{t+1}^B = M_{t+1}X_{t+1}$ buy swaps subject to quadratic adjustment cost parametrized by ζ
 - X_{t+1} : marginal value of cash (1 absent frictions)
 - Euler Equation: $\zeta s_t = E_t[M_{t+1}^B(r_{t+1} - k_t)]$
- Risk neutral measure: $E_t^*[Z_{t+1}] = E_t[M_{t+1}Z_{t+1}]$ for any Z
- After some algebra: $\zeta s_t = E_t[M_{t+1}] \text{Cov}_t^*[r_{t+1}, X_{t+1}]$

$$\zeta s_t = E_t[M_{t+1}] \text{Cov}_t^*[r_{t+1}, X_{t+1}]$$

- Bank hedging demand depends on the marginal utility-weighted covariance of interest rates with the SDF wedge
- No hedging without frictions
- When is X low?
 - Example: quadratic equity adjustment costs
 - Banks pay a target dividend as a fraction of book equity ηN_t
 - Deviating is costly. Equity issuance $x_t N_t$ incurs cost $\frac{\xi}{2} x_t^2 N_t$
 - $X_{t+1} = \frac{\eta + \frac{\xi}{2} x_{t+1}^2 + (1 + \xi x_{t+1})(1 - \eta)}{1 + \xi x_t}$ low when issuance high today
 - Can be because net worth is low 😞
 - But also, can be because investment opportunities are great 😊
 - Only indirectly related to NIMs, Equity Values, and other oft-discussed targets

$$\zeta s_t = E_t[M_{t+1}] \text{Cov}_t^*[r_{t+1}, X_{t+1}]$$

- Can be because net worth is low 😞
 - If book equity duration > 0: high rate states
 - Negative covariance → pay floating, receive fixed
- Can be because investment opportunities are great 😊
 - If markups procyclical in rates: low rate states
 - Positive covariance → pay fixed, receive floating
- Intermediary asset pricing: occasionally binding constraints are biggest source of volatility in X
 - If rate shocks cause constraints to bind, will drive most of covariance!
 - Need global model

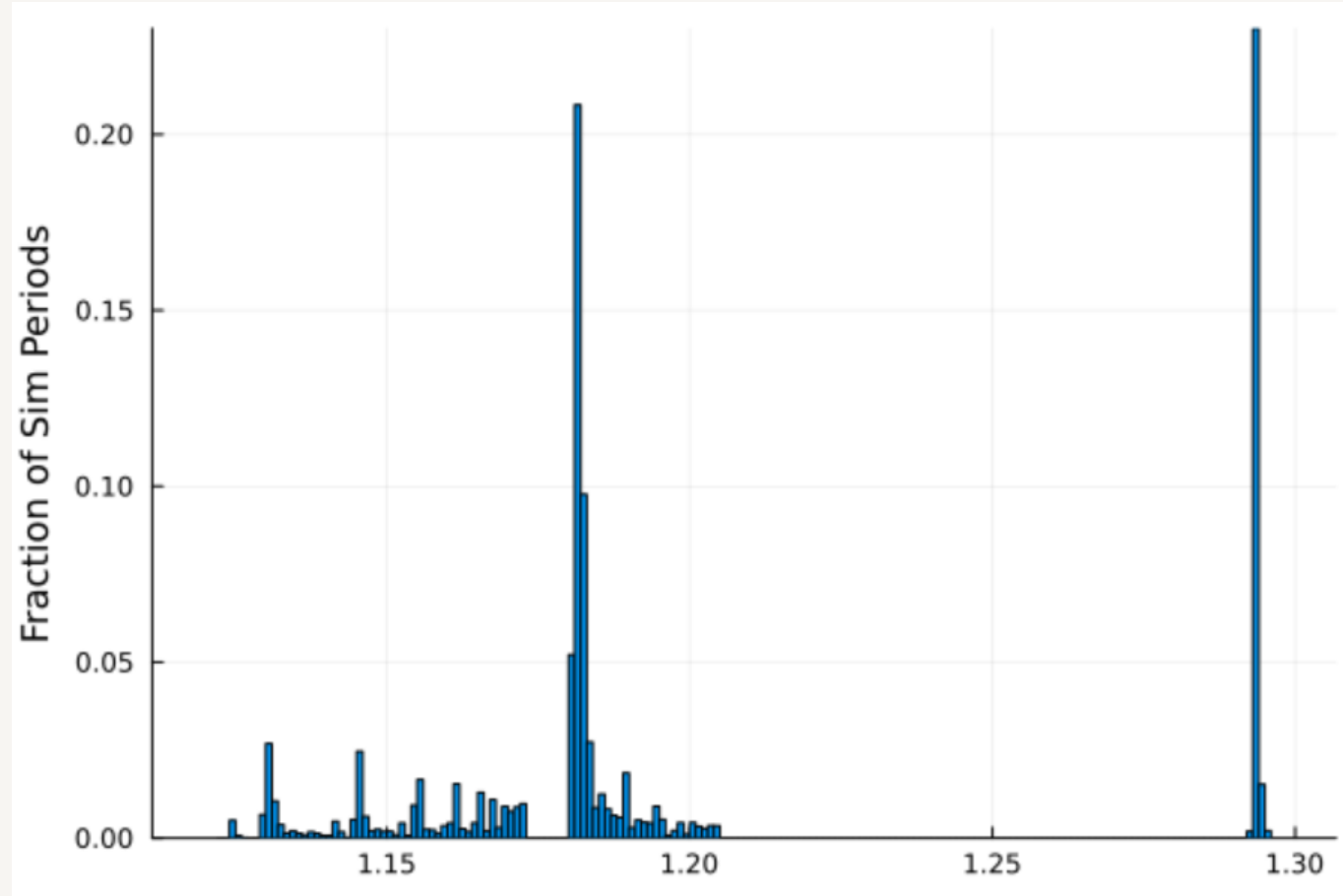
A Discrete Time Model With (a Bit) More Structure

- *Inspired by Elenev, Landvoigt, and Van Nieuwerburgh (21), Begenau, Elenev, Landvoigt (26) and Elenev and Liu (26)*
- Two aggregate shocks z (“pure” cash flow) and r (interest rates)
- Banks Subject to Equity Issuance Cost and Regulatory Constraint
- Owned by Risk-Averse Households with Other Reduced-Form Income $Y(z,r)$
- Positions in
 - Loans: Monopolistic competition with outside option
 - z & r shift both demand $A(z,r)$ and payoff $L(z,r)$ in reduced form way
 - Deposits: DSS (17) nested CES block yields limited passthrough, outflows
 - Long-Term Bonds: priced by affine term structure model (banks not marginal)
 - Swaps: as before

Three Key Objects

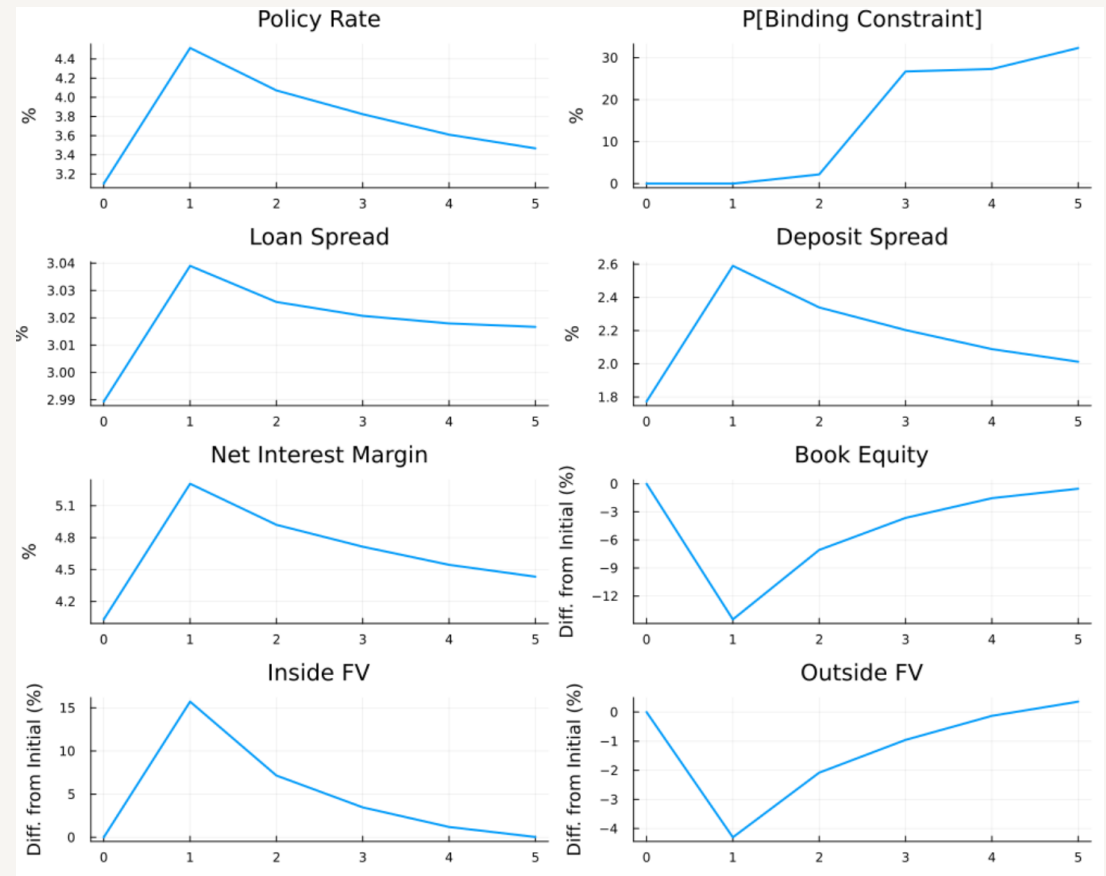
- Market Value of Equity = $N_t \times P_t^B / N_t \times P_t / P_t^B$
 - “Book” Equity N_t : Bank Net worth = Loans + Bonds + Net Swaps – Deposits
 - “Inside Franchise Value” $FV_t = \frac{P_t^B}{N_t}$
 - P_t^B : PV of dividends discounted using bank SDF
 - Inside Franchise Value $\neq 1$ because of rents, lack of entry
 - “Outside Franchise Value” $X_t = P_t / P_t^B$
 - Market Equity P_t : PV of dividends discounted using HH SDF

Constraints Make X Very Volatile



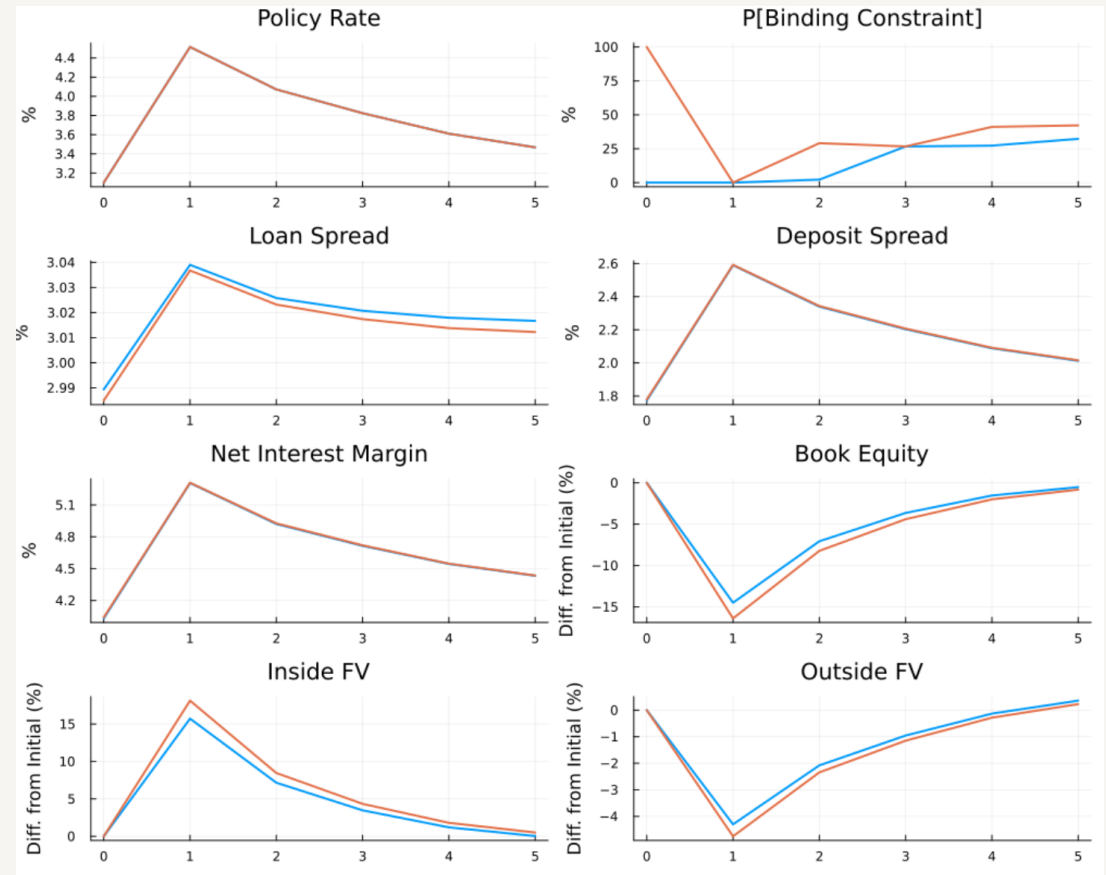
Effect of a Rate Shock

- Duration Effect on Book Equity
- Offset by Rent Effect
- Low Net Worth + Profitable Investment Opportunities → Equity Issuance → Outside FV declines



Ambiguous Equilibrium Effects of Hedging

- Duration Effect on Book Equity
- Offset by Rent Effect
- Low Net Worth + Profitable Investment Opportunities $\rightarrow$ Equity Issuance $\rightarrow$ Outside FV declines
- Ability to hedge weakens precautionary motive, constraint now binds in stochastic s.s.
- On net, makes OFV more sensitive



Conclusion

- Banks hedge rate risk to stabilize the marginal value of cash
- Even as changing rates change both cash flows and their discount rates
- Biggest threat to MV Cash stability is constraints
- Capturing the economics of the hedging problem requires a comprehensive, dynamic approach
- Excellent paper!
 - A must-read, and not just if you work in banking